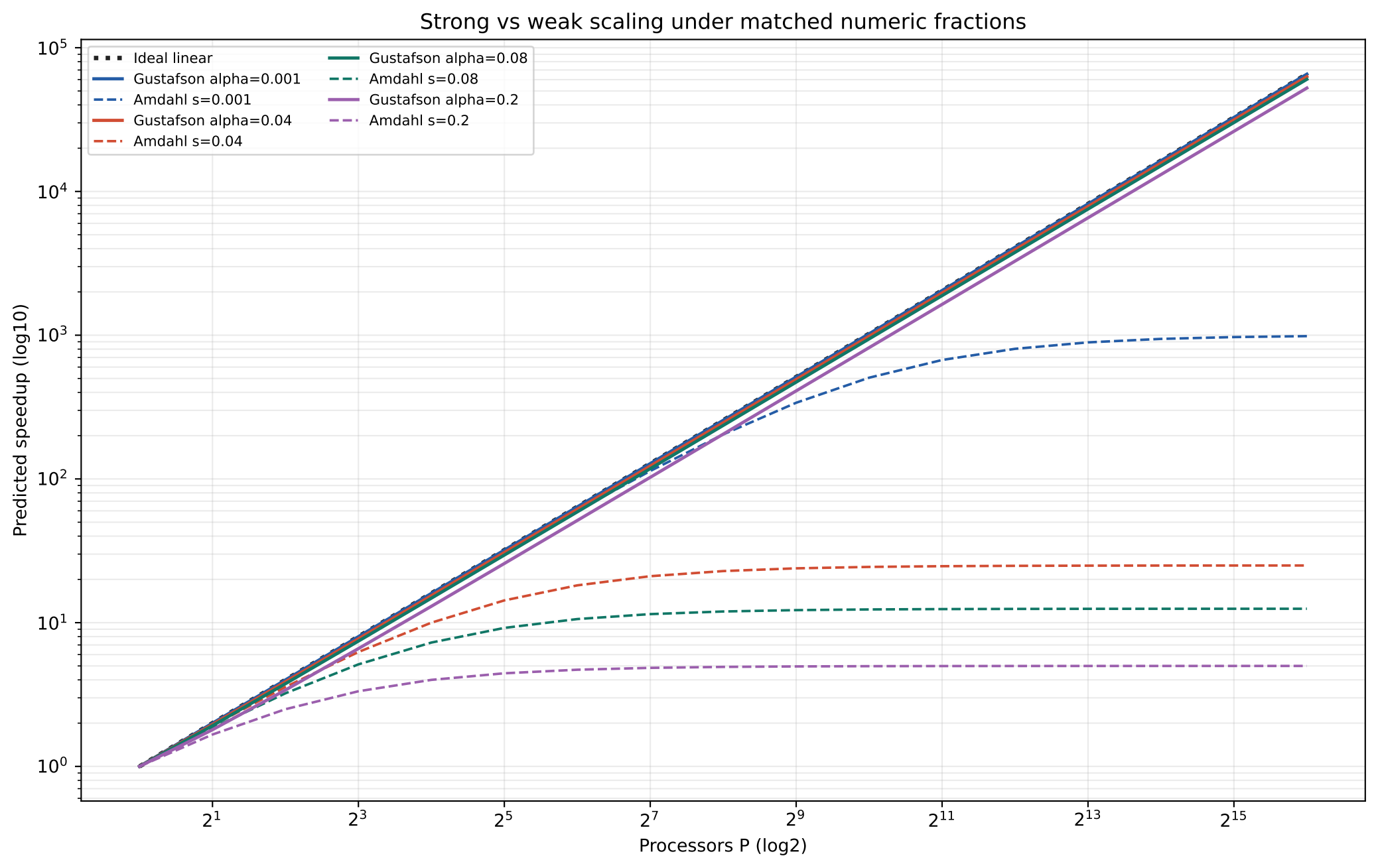
Strong vs weak scaling under matched numeric fractions
Ideal linear
Gustafson alpha=0.001
Amdahl s=0.001
Gustafson alpha=0.04
Amdahl s=0.04
Gustafson alpha=0.08
Amdahl s=0.08
Gustafson alpha=0.2
Amdahl s=0.2
Predicted speedup (log10)
Processors P (log2)
10^0
10^1
10^2
10^3
10^4
10^5
2^1
2^3
2^5
2^7
2^9
2^{11}
2^{13}
2^{15}